Unit 4 · Outcome 1 · Climate Change

Climate change projections: comparing what we saw with what the model says

KK09 VCE Environmental Science · Units 3 & 4 · Exam study summary

Digital Vector EnviroSci · single-topic mastery summary with worked good-vs-bad answers, exam traps, command words and practice.

Orientation

01 Orientation Where this dot point sits, and exactly what the examiner can ask you to do with it. ◆ VCAA study design — the examinable wording "climate change projections, including comparing observed and simulated climate, and the rating of the level of confidence (very high to very low) using the Intergovernmental Panel on Climate Change (IPCC) guidelines." Strip that sentence down and there are two examinable skills hiding inside it. First, you must be able to compare observed and simulated climate — that is, hold the real measured record next to wha

The mental model

02 The mental model One analogy that makes the whole dot point click. Think of a cricket selector's form guide A selector never says "this batter will score 80 on Saturday." That would be a prediction , and pitches, weather and bowling change everything. Instead they reason in projections : " if the wicket is dry and they bat at three, then a half-century is likely." The "if" is a scenario ; the "then" is the projection . And before they trust the form guide at all, they check it against the batter's actual past innings — that's comparing observed with

Observed vs simulated climate

03 Observed vs simulated climate The centrepiece skill: laying the real record beside the model output. Drag to orbit; toggle the two model runs. Observed climate is what we actually measured — thermometers, ice cores, tide gauges, satellites. Simulated climate is what a computer climate model (a General Circulation Model) produces when we feed it the physics of the atmosphere, oceans and land, plus a set of forcings (the things that push the energy balance: solar output, volcanoes, and greenhouse gases). The decisive experiment is to run the model twice

Hindcasting & attribution

04 Hindcasting & attribution Why running a model backwards is how we earn the right to run it forwards. Hindcasting (also called a hindcast or backtest) means running the model over a period we already have data for , then checking its output against the observations. If a model can reproduce the twentieth century — the 1991 Pinatubo cooling blip, the warming since the 1970s, the pattern of land warming faster than ocean — we gain confidence that its physics is sound, and therefore that its future projections are credible. Good hindcast → trust Simulated

Projection vs prediction

05 Projection vs prediction A distinction examiners love, because students blur it constantly. Projection A conditional "if–then" statement of how climate could evolve given a particular scenario of future emissions. Change the scenario and the projection changes. It is not a forecast of any specific year's weather. Prediction An unconditional statement that something will happen — like a weather forecast for Tuesday. Climate science deliberately avoids this for the long-term future because emissions depend on human choices we can't predict. The reason m

Scenarios & the projection fan

06 Scenarios & the projection fan Slide the emissions pathway and watch the projected warming fan out from a shared past. The IPCC feeds models a set of standard emission storylines so results are comparable worldwide. The Fifth Assessment (AR5) used Representative Concentration Pathways (RCPs) , named for the extra radiative forcing (in watts per square metre) they reach by 2100: RCP2.6 — strong mitigation Emissions fall fast, even net-negative late century. Warming held near 1.5–2 °C . The "if we act hard" branch. RCP4.5 — moderate Emissions peak then

Rating confidence the IPCC way

07 Rating confidence the IPCC way The qualitative scale — and the quantitative scale students keep confusing it with. When the IPCC makes a statement, it tags it with a confidence level from very high to very low . That level is built from two ingredients: Evidence The type, amount, quality and consistency of the evidence — observations, theory, model output, lab and field studies. More and better evidence pushes confidence up. Agreement The degree of agreement among independent studies and experts. When many lines of evidence point the same way, agreeme

Interactive: the confidence rater

08 Interactive: the confidence rater Set the evidence and the agreement, and read off the IPCC confidence — exactly how the matrix works. Agreement → Evidence: limited → robust → Evidence (type · amount · quality · consistency) Agreement among studies Very high confidence Robust evidence and high agreement — e.g. "human activity has warmed the atmosphere, ocean and land." Try the corners. Robust + high → very high (the headline climate findings). Limited + low → very low (a contested regional detail). The diagonal in between is where most careful science

Sort it out

◇ **ACTIVE RECALL**

09 Sort it out Drag each statement into the right basket. This is the exact discrimination the exam tests. Projection (conditional) If–then, tied to a scenario Prediction / forecast Unconditional "will happen" Confidence (qualitative) How robust the science is Likelihood (probability) A calibrated % outcome s

Command-word decoder

◈ **COMMAND WORDS**

10 Command-word decoder Tap a command word to see what the examiner actually wants for this dot point.

Exam traps

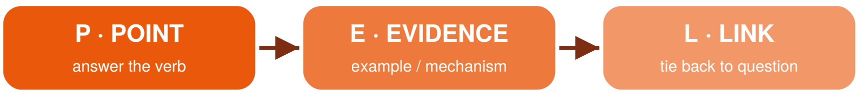
⚠ **EXAM TRAPS**

11 Exam traps The six ways students lose easy marks here. Treating a projection as a prediction Dropping the scenario condition. Always pair a projected figure with its pathway. Confusing confidence with likelihood Confidence = qualitative (evidence + agreement); likelihood = a calibrated probability. Don't write "very likely confidence." Comparing without a verdict "They're different" earns little. State which run matches the observed record and what that proves. Forgetting the human-vs-natural run The attribution argument lives in the t

P·E·L writing trainer

✎ **P·E·L WRITING**

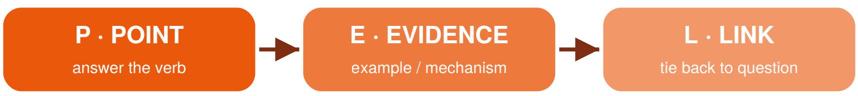
12 P·E·L writing trainer A reliable structure for the "explain / evaluate" questions on this dot point. Tap each step. P Point E Explain L Link



The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

Worked answer: weak vs strong

13 Worked answer: weak vs strong Same question, two responses. Read what lifts the strong one. The question · 4 marks Scientists ran a climate model for 1900–2020 twice — once with natural forcings only, once with natural and human forcings — and compared both with the observed temperature record. Explain how this comparison supports the conclusion that recent warming is human-caused, and comment on the confidence of that conclusion. ✕ Weak response The two model runs were different from each other. The one with humans was higher. This shows humans cause



The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

Practice & self-mark

★ **PRACTICE & SELF-MARK**

14 Practice & self-mark Five VCAA-style questions, 21 marks. Write your answer, reveal the model, then mark yourself honestly — your score saves automatically. Your score 0 / 21 Reveal a model answer to begin. Nicely done.

★ Good vs bad worked answers

Study the contrast — the same question answered two ways. The strong column shows the moves that earn the marks.

P · POINT answer the verb → **E · EVIDENCE** example / mechanism → **L · LINK** tie back to question

The P·E·L answer structure — make a Point, support with Evidence, then Link to the question.

✗ WEAK ANSWER

The two model runs were different from each other. The one with humans was higher. This shows humans cause warming . Scientists are very likely confident because the model is good. Why it loses marks: no mention of the observed record (the comparison that matters), so attribution isn't actually shown. "Very likely confident" mixes the two scales. "The model is good" isn't a justification.

✓ STRONG / MODEL ANSWER

The natural-only run failed to reproduce the observed warming after ~1970, staying roughly flat , whereas the natural-plus-human run closely matched the observed record . Because only the run including human forcings reproduces what was actually measured, the warming can be attributed to human activity rather than natural variability. Confidence is very high : the evidence is robust (multiple models, long records) and agreement among studies is high. Why it scores: centres the observed record, names both forcing sets, states the attribution explicitly, and rates confidence with the correct ingredients (evidence + agreement).

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Worked good/weak answers illustrate exam technique — always check against current VCAA marking advice.